In Excel VBA, an **array** is a powerful variable that can store a collection of values under a single name. When working with large-scale data—like processing 1,500+ problems or managing datasets across **1,048,576 rows**—arrays are significantly faster than looping through individual cells because they allow you to perform calculations in the computer's memory (RAM) rather than on the worksheet grid.

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1. Types of Arrays

Static Arrays

Use these when you know the exact number of elements you need beforehand.

- **Syntax**: `Dim myArray(1 To 5) As String`
- **Storage**: This creates 5 fixed "slots" (1 through 5).

Dynamic Arrays

Use these when the amount of data is unknown at the start, such as when you **Import** a list of varying lengths.

- **Syntax**: `Dim myDynamicArray() As Double`
- **Resizing**: Use the `ReDim` statement later in your code to set the size.
- **Preserving Data**: Use `ReDim Preserve` to change the size without losing the values already stored in the array.

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2. Multi-Dimensional Arrays

Arrays can have more than one dimension, which is perfect for representing tables (rows and columns).

- **2D Array**: `Dim salesData(1 To 10, 1 To 5)` represents 10 rows and 5 columns.
- **Accessing**: You refer to a specific "cell" in the array using `salesData(row, col)`.

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3. Speed Hack: Worksheet to Array

The most efficient way to use arrays in Excel is to dump an entire range into an array in one single line of code. This bypasses the slow process of communicating with the worksheet interface repeatedly.

```
VBA

Dim dataArray As Variant
```

```
' This one line puts thousands of rows into memory instantly
dataArray = Range("A1:B5000").Value

' Accessing the data from memory
MsgBox dataArray(1, 1)
```

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4. Key Array Functions

To manage your arrays effectively, you'll frequently use these built-in functions:

- **LBound / UBound**: Returns the "Lower Bound" and "Upper Bound" (start and end) of an array. This is essential for writing **Loops** that don't crash when the array size changes.
- **Split**: Converts a single string into an array based on a delimiter (e.g., splitting a full name into "First" and "Last" using a space).
- **Join**: The opposite of Split; it combines an array of strings into one single string.
- **Erase**: Clears the data from a static array or releases the memory used by a dynamic array.

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5. Comparison: Cells vs. Arrays

Feature	Cell-by-Cell Processing	Array Processing
Speed	Very Slow (High Overhead)	Extremely Fast (Low Overhead)
Visibility	You see the changes in real-time.	Calculations happen invisibly in RAM.
Ease of Use	Simple `Range` references.	Requires understanding of `Dim` and `Bounds`.

Best For	Small tasks (1–50 cells).	Large-scale data (1,000+ rows).

Since you've tackled over 1,500 DSA problems, you'll find that VBA arrays behave very similarly to arrays in Python or C++, though they default to being 1-indexed if you declare them as `(1 To N)`.